7. Is it right to waste helium on party balloons?

Helium is a colourless inert gas found in Group 0 of the Periodic Table.

Helium is one of the commonest elements in the Universe, second only to hydrogen. However, on Earth it is relatively rare, as shown in **Table 1**.

Gases which have a density less than air can escape the Earth's gravity and leak away into space. The density of air is 1.2 g/m^3. **Bar chart 1** shows the densities of Group 0 gases.

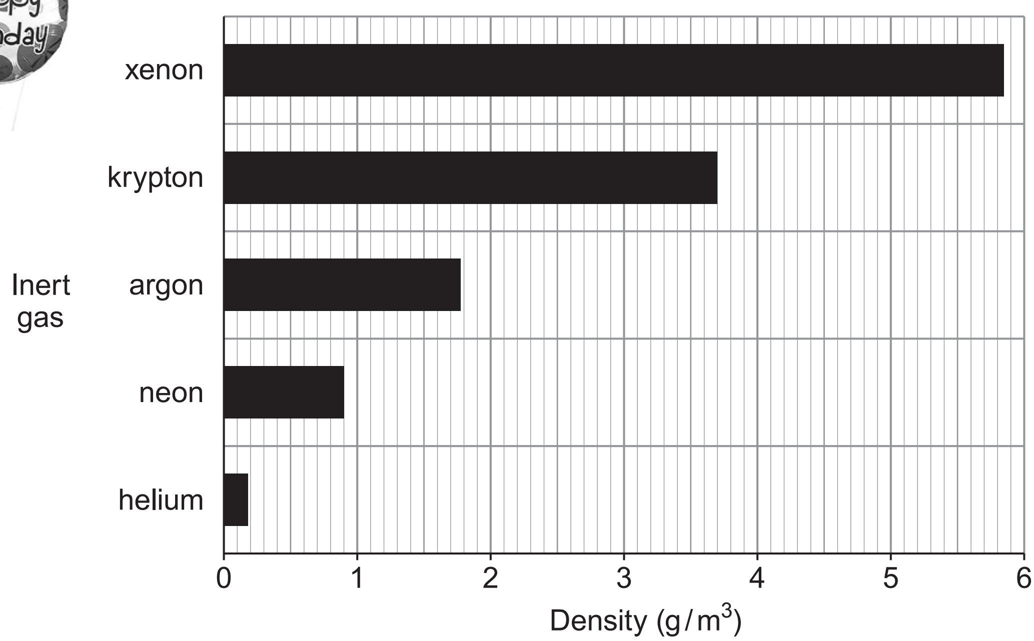
Helium has the lowest boiling point of any element. This makes it of key importance for magnets used in hospital MRI scanners, which must be super-cooled to generate the hugely powerful magnetic fields required.

Some scientists believe that because helium is a finite resource it should not be used for party balloons.

Table 1

Inert gas	Percentage in the atmosphere (%)	Melting point (°C)	Boiling point (°C)
helium	0.00052	−272	−269
neon	0.0018	−246	−246
argon	0.93	−186	−186
krypton	0.0001	−152	−152
xenon	0.000009	−111	−106

Bar chart 1



 (3410U10-1)

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(a) Answer the following questions using the information given.

(i) Tick (✓) the box next to the **most** important property that makes helium a suitable material to fill **floating** party balloons. [1]

- Helium is a gas ☐
- Helium is the second most common element in the Universe ☐
- Helium is less dense than air ☐
- Helium is colourless ☐

(ii) Tick (✓) the box next to the correct statement. [1]

- The Earth's atmosphere contains more helium than argon ☐
- The Earth's atmosphere contains more xenon than helium ☐
- The Earth's atmosphere contains more helium than krypton ☐

(iii) Tick (✓) the box next to the **best** reason for not using helium to fill party balloons. [1]

- There isn't much helium in the Earth's atmosphere ☐
- Scientists say helium shouldn't be used to fill balloons ☐
- Helium is a finite resource ☐

(iv) Tick (✓) the box next to the correct statement. [1]

- Only helium gas can leak away into space ☐
- Helium and neon gases can leak away into space ☐
- Only argon can leak away into space ☐
- All inert gases can leak away into space ☐

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(b) The table below shows the electronic structure of three Group 0 elements.

Group 0 element	Electronic structure
helium	2
neon	2,8
argon	2,8,8

Tick (✓) the box next to the statement that **best** explains why Group 0 elements are unreactive. [1]

All Group 0 elements have 2 electrons in their inner shell ☐

All Group 0 elements have 8 electrons in their outer shell ☐

All Group 0 elements have full outer shells ☐

All Group 0 elements have some full shells ☐

5

